

Metric-consistent neural networks for genomic prediction and selection: a standardized-MSE Pearson loss, affine calibration, and selection-aware evaluation

Grégory Beurier^{*,†,1}, Denis Cornet^{*,†}, Lauriane Rouan^{*,†} and David Cros^{*,†}

^{*}CIRAD, UMR AGAP Institut, Avenue Agropolis, 34398 Montpellier Cedex 5, France, [†]UMR AGAP Institut, Univ Montpellier, CIRAD, INRAE, Institut Agro, Montpellier, France

ABSTRACT Genomic prediction involves three objectives that are often conflated: predictive correlation, calibration to the phenotypic scale, and recovery of selection candidates. Neural predictors are usually trained by mean squared error (MSE), predictive ability is commonly reported as Pearson correlation, and selection ultimately depends on candidate ranking. We use the identity $\text{MSE}(z_y, z_{\hat{y}}) = 2(1 - r)$ to implement a stable differentiable Pearson objective and ask what is gained, and what is not, by aligning training with correlation. We compared MSE, Pearson, hybrid, and concordance-correlation losses for seven architectures on 101 public trait–dataset tasks from 12 panels spanning ten species. Relative to MSE, Pearson training increased raw correlation by 0.0048 (95% panel-cluster bootstrap interval 0.0008–0.0091; exact 12-panel sign-flip $p = 0.052$). Raw Pearson predictions were poorly scaled, but validation-fitted affine calibration removed this penalty while retaining the correlation gain. The corresponding change in NDCG@10 was only 0.0018 (–0.0001–0.0035), showing that improved global predictive ability did not substantially alter upper-tail recovery. Effects varied across architectures: the Transformer showed the clearest positive contrast (+0.0176), whereas the DeepGS-inspired network was negative (–0.0067), although the global interaction was inconclusive. Thus Pearson loss provides a feasible metric-consistent objective, but predictive correlation, phenotypic calibration, and selection performance should be treated as distinct targets rather than assuming that improvement in one transfers to the others.

KEYWORDS

genomic prediction
genomic selection
deep learning
loss functions
calibration

INTRODUCTION

Genomic selection (Meuwissen *et al.* 2001) predicts the genetic merit of candidate individuals from genome-wide markers, and has become central to modern plant and animal breeding (Heffner *et al.* 2009; Jannink *et al.* 2010; Desta and Ortiz 2014). The dominant parametric baseline, genomic best linear unbiased prediction (GBLUP)/ridge-regression BLUP (VanRaden 2008; Endelman 2011), is increasingly complemented by machine-learning and deep-learning predictors (Ma *et al.* 2018; Wang *et al.* 2023; Montesinos-López *et al.* 2021b,a). Whether deep learning improves on GBLUP remains dataset-dependent: recent comparisons

find that neither systematically dominates, and that outcomes hinge on tuning, architecture and the choice of evaluation metric (Montesinos-López *et al.* 2025).

That last point is the subject of this paper. Genomic predictors are trained, by near-universal default, to minimize the mean squared error (MSE). But they are *assessed* with the Pearson correlation r between predicted and observed phenotypes — the “predictive ability” reported in essentially every genomic prediction study (Crossa *et al.* 2010) — and they are *used* to rank candidates and select the best ones. There is therefore a systematic mismatch between the training objective and one of the principal metrics used to evaluate genomic prediction. More importantly, neither MSE nor Pearson correlation is identical to the ranking problem ultimately faced in selection. Waldmann (2019) showed that selecting a model by r can disagree with selecting it by MSE or R^2 .

Manuscript compiled: Monday 5th October, 2026

[†]Corresponding author: CIRAD, UMR AGAP Institut, Avenue Agropolis, 34398 Montpellier Cedex 5, France. E-mail: gregory.beurier@cirad.fr

Ornella *et al.* (2014) argued that r is a poor guide to performance in the tails of the distribution, where selection actually operates, and evaluated the recovery of the best $\alpha \in \{10, \dots, 40\}\%$ of individuals. Blondel *et al.* (2015) reframed genomic selection as a ranking problem, introduced NDCG from information retrieval as an accuracy measure, and reported that r correlates poorly with ranking quality.

If correlation is the target, why not train for it directly? A recent convolutional architecture for genomic selection states plainly that “Pearson correlation cannot be set as the loss function” and substitutes a family of L_p losses (Xie *et al.* 2024). We show this premise is false. The Pearson correlation is a loss — a standardized MSE — and training with it is both well posed and numerically stable. The remaining objection to a correlation loss, that it ignores scale and offset, is real but easily and provably remedied by an affine recalibration applied after training.

The empirical question therefore has a simple three-stage structure: compare the usual MSE-trained raw predictions with Pearson-trained raw predictions to measure correlation alignment, then calibrate the Pearson outputs on validation data to recover phenotypic scale, and finally evaluate separately whether correlation-aligned training changes upper-tail selection. Figure 4 carries this structure through to the headline result.

Contributions:

1. **A standardized-MSE Pearson loss (Section The Pearson loss is a standardized MSE).** We give the exact identity $\text{MSE}(z_y, z_{\hat{y}}) = 2(1 - r)$ and use it as a stable, differentiable training objective, verified numerically (to single-precision tolerance for the loss; to float64 machine precision for the calibration identity).
2. **Affine calibration with a closed form (Section Affine calibration restores scale without changing rank).** We prove that the optimal post-hoc affine map yields residual error $\sigma_y^2(1 - r^2)$ on the fitting sample and restores scale and offset while preserving r and rankings whenever the fitted slope is positive.
3. **Mini-batch analysis (Section Mini-batch versus global Pearson).** We show the mini-batch Pearson loss is a biased surrogate for the global objective and quantify the effect of batch size empirically.
4. **A reproducible, selection-aware benchmark (Sections Materials and Methods–Results).** Across 101 trait–datasets from 12 panels/sources spanning 10 species, with architecture, splits and tuning held fixed so that only the loss changes, we report both predictive and selection-oriented metrics, a confirmatory mixed-model analysis, and a meta-analysis of *when* correlation-consistent training helps. All data are public and all code, splits and tuned configurations are released.

We do not claim that a Pearson loss is uniformly superior to MSE; that would be false. We provide a principled, reproducible framework for training, calibrating and evaluating neural genomic predictors by treating predictive correlation, phenotypic scale and selection as related but distinct objectives.

THEORY

Throughout, $y, \hat{y} \in \mathbb{R}^n$, with sample means $\bar{y}, \bar{\hat{y}}$ and population variances $\sigma_x^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2$. The Pearson correlation is $r = r(y, \hat{y}) = \frac{1}{n} \sum_i z_{y,i} z_{\hat{y},i}$ where $z_{x,i} = (x_i - \bar{x}) / \sigma_x$ is the standardized variable.

The Pearson loss is a standardized MSE

Proposition 1. For non-degenerate $y, \hat{y}$ (i.e. $\sigma_y, \sigma_{\hat{y}} > 0$),

$$\text{MSE}(z_y, z_{\hat{y}}) = \frac{1}{n} \sum_{i=1}^n (z_{y,i} - z_{\hat{y},i})^2 = 2(1 - r).$$

Consequently $1 - r = \frac{1}{2} \text{MSE}(z_y, z_{\hat{y}})$, and minimizing the standardized MSE is exactly maximizing the Pearson correlation.

Proof. Expanding the square, $\frac{1}{n} \sum_i (z_{y,i} - z_{\hat{y},i})^2 = \frac{1}{n} \sum_i z_{y,i}^2 - 2 \frac{1}{n} \sum_i z_{y,i} z_{\hat{y},i} + \frac{1}{n} \sum_i z_{\hat{y},i}^2 = 1 - 2r + 1 = 2(1 - r)$, because $\frac{1}{n} \sum_i z_{x,i}^2 = 1$ by construction and $\frac{1}{n} \sum_i z_{y,i} z_{\hat{y},i} = r$. $\square$

This gives the loss $\mathcal{L}_P(\hat{y}, y) = 1 - r(\hat{y}, y)$, implemented directly with a small ε guarding the standard-deviation denominators and numerically equivalent to $\frac{1}{2} \text{MSE}(z_y, z_{\hat{y}})$. It is differentiable everywhere $\sigma_{\hat{y}} > 0$; we verify the gradient against finite differences and verify the identity to single-precision tolerance in Section Numerical validation. The negative-correlation loss $-r$ differs from $\mathcal{L}_P$ by a constant and has identical gradients.

Affine calibration restores scale without changing rank

Correlation is invariant to strictly increasing affine transformations of $\hat{y}$: a model trained on $\mathcal{L}_P$ controls the *shape* of the prediction but not its scale or offset. This is corrected exactly and cheaply.

Proposition 2. Let $p \in \mathbb{R}^n$ be raw predictions and $r = r(y, p)$. The affine map $\tilde{y} = a p + b$ minimizing $\frac{1}{n} \sum_i (a p_i + b - y_i)^2$ has

$$a^* = \frac{\text{cov}(y, p)}{\text{var}(p)}, \quad b^* = \bar{y} - a^* \bar{p}, \quad \text{MSE}_{\min} = \sigma_y^2 (1 - r^2).$$

If $a^* > 0$, the map is strictly increasing and hence preserves r , the Spearman correlation, and every top-k/ranking quantity, while setting the calibration slope to 1 and intercept to 0.

Proof. Ordinary least squares for a single predictor gives a^*, b^* as stated; the minimized residual variance of simple linear regression is $\sigma_y^2(1 - r^2)$. A strictly increasing transform preserves the order of p and leaves correlation-type and rank-based statistics unchanged. $\square$

We fit (a^*, b^*) on held-out validation predictions (never on the test set) and apply them to the test predictions. Isotonic regression provides a monotone non-linear alternative when the raw-vs-true relationship is monotone but curved. A positive affine map preserves r ; a negative slope changes its sign, and nonlinear isotonic calibration can change its magnitude. Calibration can nonetheless improve RMSE, bias and slope (Figure 2; Section Calibration).

Mini-batch versus global Pearson

Proposition 3. For a partition of the data into batches $B_1, \dots, B_K$, the batch-averaged loss $\frac{1}{K} \sum_k (1 - r_{B_k})$ is not equal to $1 - r_{\text{global}}$ in general, because r is a non-linear (ratio-of-moments) functional of the data. Because the functional is nonlinear, neither equality nor a monotonic relation between batch size and discrepancy is guaranteed without further assumptions.

A full-batch evaluation of $\mathcal{L}_P$ optimizes the global objective exactly; a large batch is an approximation, and smaller batches optimize a noisier surrogate. A supporting experiment quantifies the practical effect over the tested range.

Multi-trait and group-aware variants

The loss extends naturally to T traits, $\mathcal{L} = \frac{1}{T} \sum_{t=1}^T (1 - r_t)$ (or a weighted sum $\sum_t w_t (1 - r_t)$), and to a group-aware objective $\mathcal{L} = \frac{1}{G} \sum_{g=1}^G (1 - r_g)$ where g indexes environments, years, families or sites — aligning training with the within-environment correlation that multi-environment breeding programs actually care about. As in Proposition 3, neither variant equals the pooled global correlation.

MATERIALS AND METHODS

Datasets

We use three public sources, for a total of 101 trait–dataset combinations across 12 panels/sources spanning 10 species. (i) *EasyGeSe* (Quesada-Traver *et al.* 2025): a benchmark of ten species (barley, common bean, lentil, maize, eastern oyster, pig, loblolly pine, rice, soybean and wheat) providing genotypes, phenotypes and fixed cross-validation partitions; the maize set is the global genotype-by-environment competition panel (Washburn *et al.* 2025). (ii) The CIMMYT wheat panel (Cossa *et al.* 2010): 599 lines, 1,279 DArT markers and grain yield in four environments, analyzed as four outcomes. (iii) *SoyNAM* (Xavier *et al.* 2016): ~5,500 recombinant inbred lines in 40 biparental families, analyzed for four agronomic traits. Genotypes are coded as allele dosages; preprocessing (unsupervised, no phenotype) applies a minor-allele frequency filter (≥ 0.01), mean imputation, and a top-variance cap at 20,000 markers for the seven panels whose marker count exceeded 20,000 (barley, lentil, maize, oyster, pig, rice and soybean). Table 1 and Figure 3 summarize the data.

Models

The parametric baseline is GBLUP with REML-estimated variance components, implemented in the spectral basis of the linear genomic kernel $K = XX^T/p$ and validated against the rrBLUP package (Endelman 2011) to floating-point precision (cross-implementation correlation 1.0000). We also include ridge regression as a linear baseline. The neural predictors are a multilayer perceptron (MLP), a one-dimensional convolutional network over ordered markers (1D-CNN), a light Transformer, and protocol-matched, literature-inspired adaptations of DeepGS (Ma *et al.* 2018), DNNGP (Wang *et al.* 2023), PNNGS (Xie *et al.* 2024), and SoyDNGP (Gao *et al.* 2023). They retain published motifs under common dosage inputs, preprocessing and evaluation and are not exact reproductions. One configuration was selected per dataset registry key and architecture under MSE and frozen across losses. EasyGeSe and CIMMYT transfer a representative-trait configuration within panel; each SoyNAM phenotype is a separately tuned key. This may favor MSE.

Loss functions

We compare MSE; the standardized-MSE Pearson loss $1 - r$ (Proposition 1); a hybrid $(1 - r) + \lambda \text{MSE}$ with $\lambda = 0.1$; and a concordance-correlation loss $1 - \rho_c$ (Lin 1989), which penalizes correlation, scale and offset jointly. The negative-Pearson and explicit standardized-MSE forms are included in the numerical checks.

Calibration

For every model and loss we report predictions under three post-hoc maps: raw (identity), affine (Proposition 2) and isotonic. Calibrators are fit on validation predictions only.

Cross-validation

We use the fixed EasyGeSe partitions (5 folds $\times$ 2 repeats = 10 folds per trait) and repeated K -fold for wheat and SoyNAM. Within each training fold an inner 80/20 split provides early-stopping and calibration data; markers are standardized with training statistics only; targets are standardized with training statistics for the neural networks and predictions are returned in the original units. Neural networks early-stop on the validation value of the loss being trained. Test folds are excluded from model fitting, early stopping and calibration. Primary HPO used the full tuning phenotype before outer CV, so absolute performance is not nested. A separate CIMMYT sensitivity selected MSE and Pearson configurations from identical eight-candidate banks inside five outer folds using environment-1 outer-training observations, then evaluated four environments with three seeds.

Evaluation metrics

Predictive: Pearson r , Spearman ρ , RMSE, MAE, R^2 , mean bias, and the calibration slope and intercept (ideal 1 and 0). Upper-tail ranking, at intensities $k \in \{5, 10, 15, 20\}\%$: top- k overlap, precision@ k , recall@ k , NDCG@ k (Järvelin and Kekäläinen 2002), selection differential, mean phenotype of the selected, and relative efficiency (achieved upper-tail differential divided by the oracle differential). Larger phenotypes are always targeted, so these are not interpreted as breeding gain for traits where lower or intermediate values are desirable.

Statistical Analysis

Folds were first averaged within each task, architecture and loss before forming paired loss-minus-MSE contrasts. Architectures were averaged within task and the primary estimand weighted the 12 panels equally; a trait-weighted analysis was a sensitivity check. Intervals used 20,000 panel-cluster bootstrap replicates and an exact sign-flip test reversed whole panels. A corroborative mixed model used loss, architecture, their interaction, panel blocking and a random task intercept. Raw Pearson was primary, raw NDCG@10 secondary, and affine RMSE was normalized by test-fold phenotype SD. The one-panel nested-HPO sensitivity covered MLP, CNN and Transformer only and was analyzed descriptively. The framework is released as the Python package *cggp*, built on PyTorch (Paszke *et al.* 2019), scikit-learn (Pedregosa *et al.* 2011) and NumPy (Harris *et al.* 2020), with the neural networks trained using the Adam optimizer (Kingma and Ba 2015). The main campaign used one initialization per fold; the nested wheat sensitivity used three. Analysis scripts validate 90,900 primary and 1,080 nested rows before generating tables and figure source data.

Data availability

All datasets are public: EasyGeSe (Zenodo, DOI 10.5281/zenodo.15348871), the CIMMYT wheat panel (BGLR R package (Pérez and de los Campos 2014)) and SoyNAM (SoyNAM R package). Code, fixed cross-validation partitions, tuned configurations and a recorded runtime manifest are released as the *cggp* package, with source at <https://github.com/GBeurier/gs-loss> (a versioned archive will be deposited at Zenodo, DOI to be assigned). All reported benchmark datasets are public. Supporting Information documents architecture adaptations, paired contrasts, missingness, HPO provenance, calibration, mean ranks and the nested wheat sensitivity; machine-readable source tables accompany the figures.

RESULTS

Numerical validation

Across 80 simulated cases, the implemented Pearson loss, $1 - r$, and one half of the standardized MSE agreed to a maximum absolute difference of 5.0×10^{-8} (mean 7.4×10^{-9}). The affine residual identity in Proposition 2 agreed to 1.2×10^{-15} over 200 cases. Both the Pearson and concordance losses passed automatic gradient checks, and $-r$ and $1 - r$ had identical gradients. These checks establish numerical equivalence, not empirical superiority.

Headline result: correlation, calibration and selection separate

Figure 4 summarizes the practical sequence. Relative to the MSE-trained raw baseline, Pearson-trained raw predictions gained 0.0048 in correlation but increased normalized RMSE by 0.346 (95% panel-cluster bootstrap interval $[0.1965, 0.5675]$), because correlation training does not identify scale or offset. Validation-fitted affine calibration retained nearly all of the correlation gain ($+0.0047$) and brought normalized RMSE back to the MSE baseline (-0.0033 , $[-0.0062, 0.0000]$). Thus the headline is not that Pearson replaces MSE on every criterion: it slightly improves the target correlation, a separate calibration step is needed for phenotypic-scale predictions, and the decision-level effect must be checked directly.

The final grid contained 90,900 rows: seven neural architectures, four losses, 101 tasks, ten test folds and three calibration maps, plus GBLUP and ridge baselines. All architectures covered the same 101 tasks. After folds were averaged, the panel-balanced Pearson-minus-MSE contrast for raw test correlation was $\Delta r = +0.0048$ (95% panel-cluster bootstrap interval $[0.0008, 0.0091]$). Seventy-eight of 101 task-level pooled contrasts were positive. An exact test that flipped the signs of all tasks within each panel gave $p = 0.052$, appropriately reflecting that only 12 panels, not 707 independent experiments, were observed. Weighting every trait equally instead produced $+0.0101$ ($[0.0028, 0.0158]$), showing that the estimate is sensitive to the many traits contributed by rice, pine and wheatG.

The hybrid and CCC losses gave panel-balanced Δr values of $+0.0055$ ($[0.0010, 0.0100]$) and $+0.0078$ ($[0.0022, 0.0139]$), respectively. Figure 4D reports all seven architecture-specific Pearson contrasts and intervals. The Transformer has the largest positive estimate ($+0.0176$), while DeepGS is a negative counterexample (-0.0067); PNNGS and SoyDNGP are positive but uncertain, MLP shows a smaller positive estimate, and DNNGP and CNN lie near zero. The apparent spread is descriptive: the global loss-by-architecture test is inconclusive ($\chi^2_{12} = 12.40$, $p = 0.414$). The four named networks are literature-inspired adaptations, not exact reproductions of the published systems.

Ranking and scale

Pearson training changed panel-balanced raw NDCG@10 by only $+0.0018$ ($[-0.0001, 0.0035]$) and upper-tail overlap by $+0.0033$ ($[-0.0010, 0.0069]$). CCC gave the largest, still small, NDCG contrast ($+0.0034$, $[0.0015, 0.0056]$). These quantities recover the highest observed phenotypes; because trait desirability was not harmonized, they must not be read as realized breeding gain for traits where lower values are preferred. Ridge and GBLUP retained the best mean ranks for both Pearson and NDCG@10. The weak ranking response is therefore an empirical result in its own right: a measurable gain in global predictive ability did not substantially change which candidates would be recovered by the upper-tail rule used here.

Affine-calibrated NRMSE, defined as RMSE divided by the test-fold phenotype standard deviation, was essentially unchanged by Pearson training (-0.0011 , $[-0.0036, 0.0015]$) or the

hybrid loss (-0.0018 , $[-0.0041, 0.0006]$). CCC produced -0.0028 ($[-0.0054, -0.0003]$), a very small normalized-error reduction. Absolute RMSE values were not pooled across traits because their measurement units differ.

Calibration

Across Pearson-trained networks, mean RMSE fell from 10.54 to 7.99 after affine calibration and the median calibration slope moved from 0.918 to 0.957. These raw-unit averages are descriptive only. A positive affine map preserves ranks, but the implementation used the unconstrained least-squares slope estimated on validation data. Among 7,014 comparable fold cells, 12 (0.17%) therefore reversed the sign of test correlation. All correlation and upper-tail claims in this paper consequently use raw predictions; affine predictions are used only for error and calibration outcomes.

Loss-specific HPO and seed sensitivity

The nested sensitivity contained 1,080 prediction rows. It was run only for MLP, CNN and Transformer; DeepGS, DNNGP, PNNGS and SoyDNGP therefore have no loss-specific nested-HPO result, rather than an omitted result (Figure 6). Across the three tested architectures, loss-specific Pearson tuning changed raw r by $+0.0093$ on average. Estimates were $+0.0286$ for the Transformer, $+0.0038$ for the MLP, and -0.0044 for the CNN. Corresponding NDCG@10 contrasts were $+0.0069$, $+0.0071$, and -0.0060 . CNN's mean Δr moved from -0.0297 to -0.0041 to $+0.0207$ across seeds 0–2; Transformer remained positive ($+0.0246$, $+0.0259$, $+0.0352$). Thus objective-specific tuning supports the Transformer signal but also demonstrates initialization uncertainty that the single-seed main campaign cannot quantify.

DISCUSSION

Objective alignment, calibration and selection

The benchmark establishes a three-step workflow: align the neural training objective with predictive correlation, restore phenotypic scale with a separate calibration step, and then evaluate the selection decision directly. The algebraic result is exact: minimizing standardized MSE maximizes Pearson correlation. Empirically, once all seven networks are evaluated on the same 101 tasks and panels are equally weighted, Pearson training improves r by about five thousandths. The cluster-bootstrap interval is positive, but an exact 12-panel sign-flip test lies at the conventional threshold. The raw Pearson outputs pay a large scale penalty; validation-fitted affine calibration removes it while leaving the correlation gain essentially intact. NDCG changes are smaller still. Thus the benchmark identifies both what correlation-aligned training can deliver and what it cannot be assumed to deliver.

Trait weighting matters. A trait-weighted analysis gives twice the panel-balanced effect because rice, pine and wheatG contribute many phenotypes. Neither estimand is intrinsically wrong: the first describes an average listed trait and the second an average panel. We use panel balance for the headline to avoid allowing heavily phenotyped resources to determine a cross-panel claim, and report both for transparency.

CCC produced slightly larger pooled improvements in correlation and upper-tail ranking, consistent with its joint treatment of correlation, scale and offset. Pearson is therefore used here as the cleanest case of metric-consistent correlation training, rather than because it uniformly outperformed the alternative objectives.

Architecture heterogeneity is suggestive, not universal

Objective sensitivity may be architecture-dependent: the Transformer repeatedly shows the largest positive estimate, including after loss-specific nested tuning. MLP gains are small, CNN is near zero after panel balance, and the DeepGS-inspired network becomes worse under Pearson training. This counterexample rules out a universal optimization advantage. At the same time, the omnibus loss-by-architecture interaction is not significant, so the apparent ordering should be treated as hypothesis-generating rather than as a formal taxonomy of networks. Possible mechanisms include different curvature, normalization and early-stopping behavior, but the present benchmark does not identify them.

The four named systems are protocol-matched, literature-inspired adaptations. DeepGS retains a marker convolution/dense motif but adds batch normalization; DNNGP is genotype-only and uses PCA rather than reproducing its complete multi-omic setting; PNNGS retains parallel residual kernels but changes the stem, pooling and validation protocol; and SoyDNNGP converts dosage vectors to a 96×96 one-channel representation rather than the published $206 \times 206 \times 3$ encoding. Their results evaluate whether these motifs respond to the loss under one common protocol, not whether the original papers would reproduce identically.

Correlation, calibration and upper-tail recovery

Correlation is invariant to positive affine transformations, making scale calibration a separate problem. Validation-fitted affine calibration reduced normalized error in aggregate, but unconstrained least squares occasionally estimated a negative slope and reversed rankings. A production implementation should constrain the slope to be positive or fall back to an intercept-only map. We avoided this ambiguity by using raw outputs for correlation and ranking and affine outputs only for scale outcomes.

The upper-tail metrics answer a narrower question than biological selection. They quantify recovery of the largest observed phenotypes. Some disease, maturity and moisture traits favor lower or intermediate values, and the public metadata did not provide a uniform desirability direction. Consequently these metrics cannot establish breeding utility without trait-specific economic weights and target directions. Their weak improvement nevertheless shows that a small gain in global correlation need not translate into a substantial change in the selected set, consistent with earlier ranking-based critiques (Ornella *et al.* 2014; Blondel *et al.* 2015).

What the HPO comparison does and does not show

The main experiment is a controlled shared-configuration ablation. For each panel and architecture, a configuration selected under MSE on one representative trait was transferred to all traits and frozen across losses. This isolates the training objective conditional on that configuration, but it can favor MSE and uses the tuning phenotype before outer evaluation; it is not nested performance estimation. The separate CIMMYT experiment selects MSE and Pearson configurations inside each outer-training set from identical candidate banks. Its Transformer result is directionally consistent with the main analysis, but the experiment tunes on environment 1, transfers to four correlated wheat environments, and covers only three architectures. It is a sensitivity analysis, not a second general benchmark.

Limitations and future work

The main campaign uses one neural initialization per fold. The three-seed wheat analysis shows that this is a material limitation, especially for CNN. Future work should repeat all task-

architecture cells over seeds, nest tuning within every outer fold and phenotype, and record the full HPO trial ledger rather than only selected configurations. With only 12 panels, resampling uncertainty also remains coarse even though folds and architectures are not treated as independent replicates.

The marker filter, imputation and variance cap were fitted transductively on the full genotype matrix but without phenotypes; fold-specific standardization was then fitted on training individuals. Binary, ordinal and continuous phenotypes were analyzed together for breadth, although Gaussian calibration and RMSE have different meanings across these scales. Constant or nearly constant test folds also generated missing correlations, concentrated in rice seed color. These limitations motivate stratified future benchmarks.

Finally, the benchmark compares objective functions, not every possible neural architecture, and linear baselines remain strongest by mean rank. The practical recommendation is therefore to use Pearson or CCC as paired objective-specific ablations when correlation is the operational endpoint; retain MSE and linear baselines; calibrate separately when phenotypic scale matters; and verify the actual trait-specific selection decision rather than inferring it from r .

ACKNOWLEDGMENTS

The authors acknowledge the partners of the Bana+ project for their contribution to and support of this work. We express our sincere gratitude to the researchers, consortia and package maintainers who created and publicly shared the datasets used in this benchmark: the EasyGeSe team (Quesada-Traver *et al.* 2025), the CIMMYT wheat breeding program and the BGLR team (Crossa *et al.* 2010; Pérez and de los Campos 2014), and the SoyNAM project investigators and package developers (Xavier *et al.* 2016). The authors also acknowledge the symbolic and institutional support of the collective research initiative Camille Noûs (Laboratoire Cogitamus).

STUDY FUNDING

This work was supported by the Bana+ project, funded by the French Ministry of Agriculture under the PARSADA programme.

CONCLUSION

A standardized-MSE Pearson loss aligns neural-network training with the predictive-ability metric used throughout genomic prediction; because correlation is scale-free, a separate calibration step or a scale-aware objective is needed when calibrated phenotypic values matter. Across 101 tasks the average benefit was small, sensitive to panel weighting and not uniform across architectures. The Transformer gave the clearest descriptive signal, DeepGS a negative counterexample, and no neural loss displaced GBLUP or ridge on mean rank. The results nevertheless provide a practical framework for separating objective alignment, phenotypic calibration and decision-level evaluation in neural genomic prediction; Pearson or CCC should be viewed as objective-specific alternatives to test alongside MSE rather than as universal replacements.

LITERATURE CITED

- Blondel, M., A. Onogi, H. Iwata, and N. Ueda, 2015 A ranking approach to genomic selection. *PLoS ONE* 10: e0128570.
- Crossa, J., G. de los Campos, P. Pérez, D. Gianola, J. Burgueño, *et al.*, 2010 Prediction of genetic values of quantitative traits in plant breeding using pedigree and molecular markers. *Genetics* 186: 713–724.

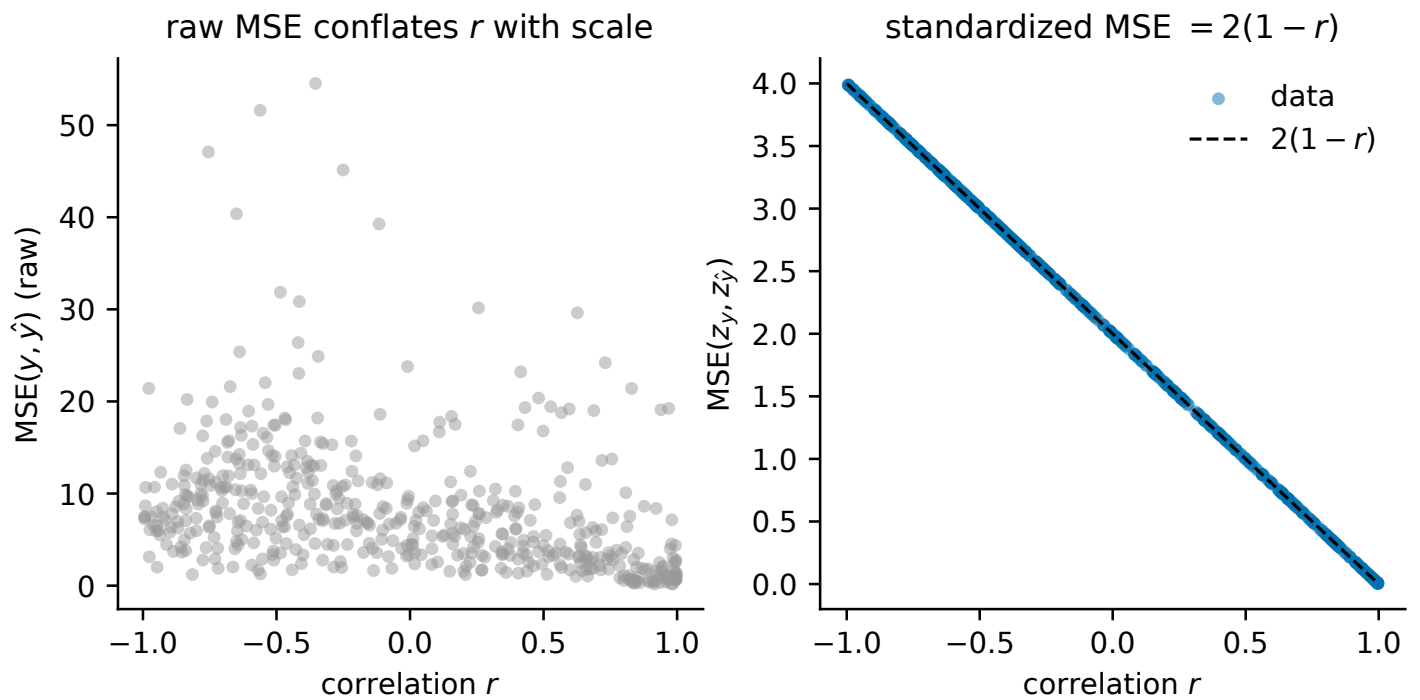


Figure 1 Geometry of the standardized-MSE / Pearson identity (Proposition 1).

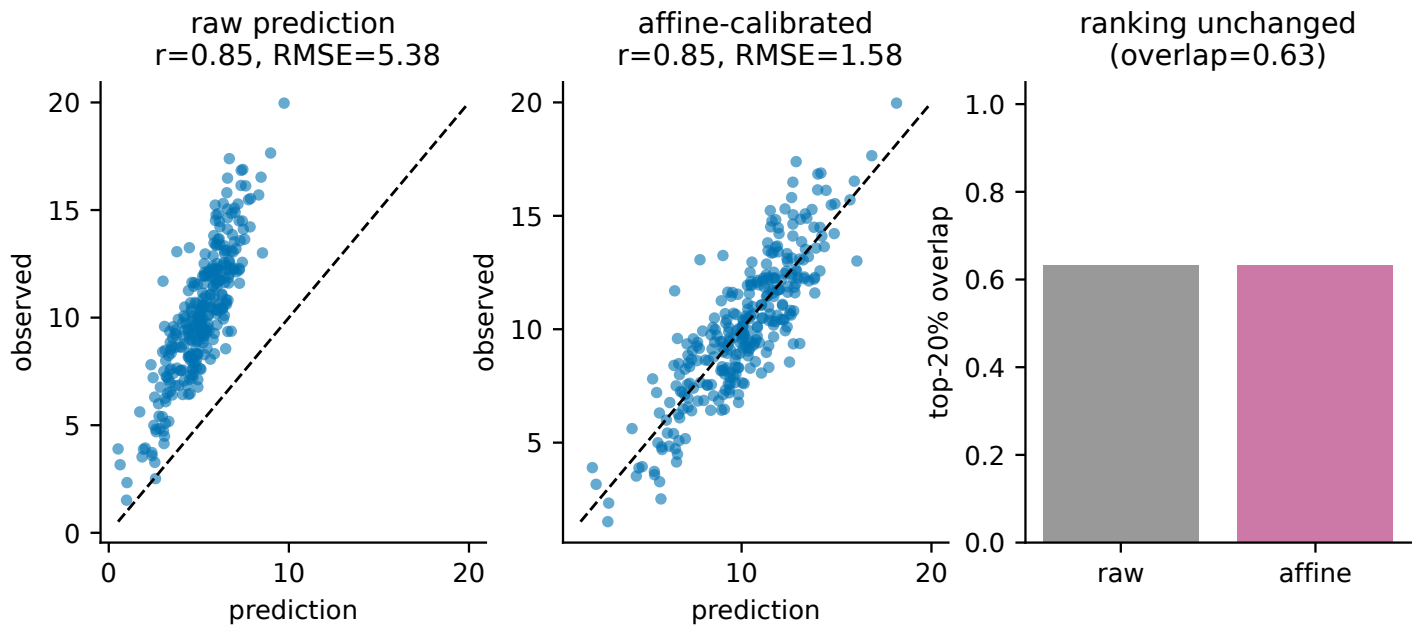


Figure 2 Controlled simulation of Proposition 2: a well-correlated but mis-scaled prediction (left) is corrected by affine calibration (middle) — RMSE falls while Pearson r is unchanged — and the top-20% selection overlap is identical before and after (right).

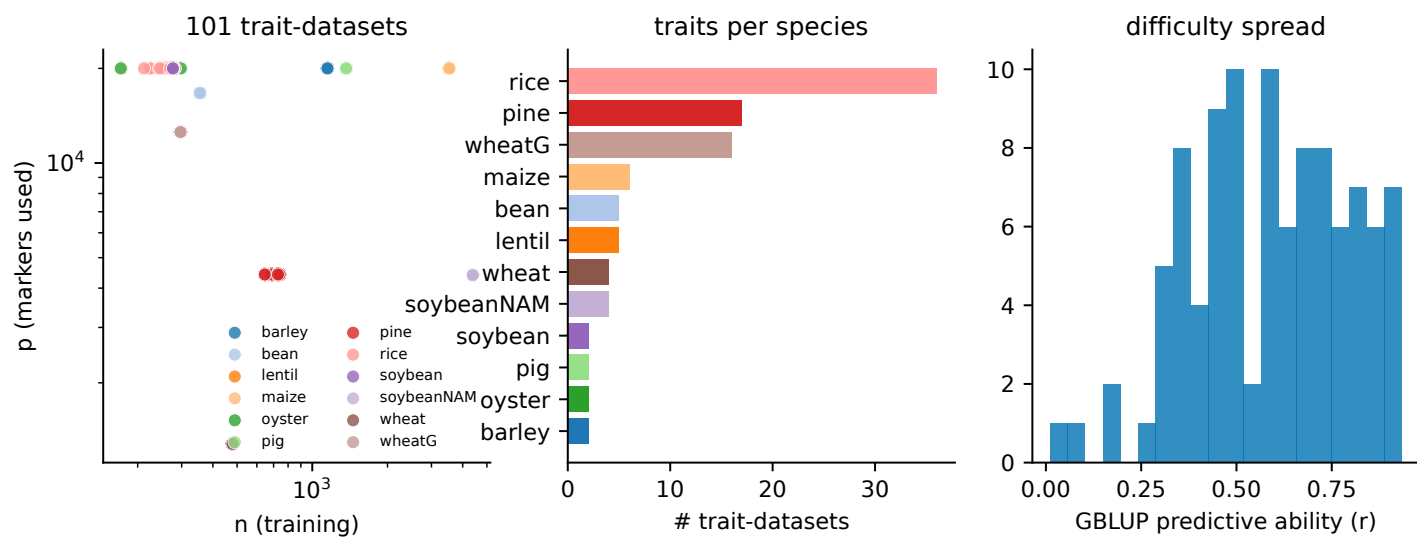
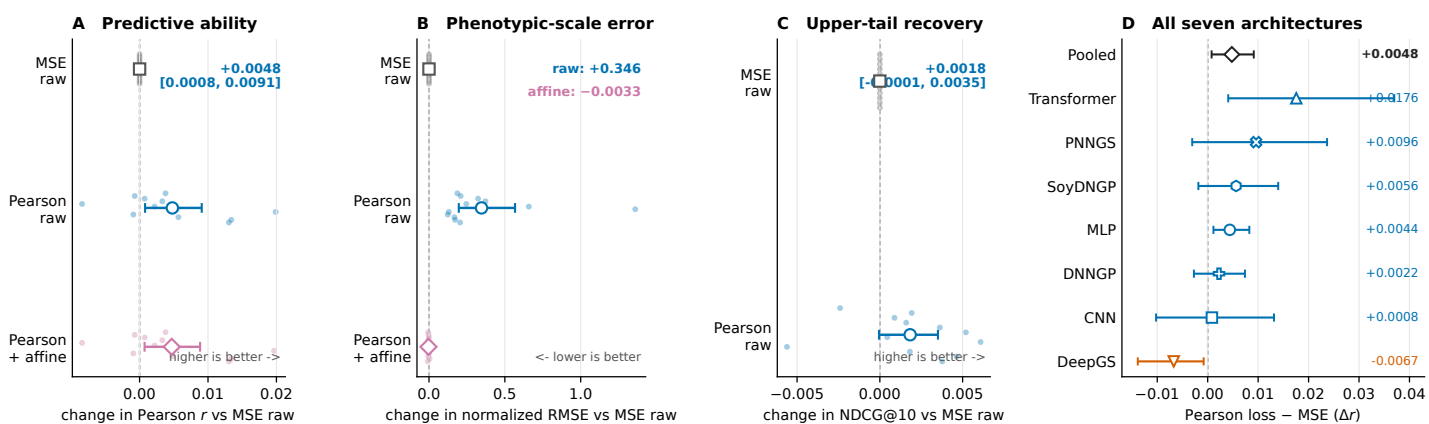


Figure 3 Dataset map: size, dimensionality and predictability across species.

Pearson loss separates correlation, calibration, and selection effects



Small points (A-C): 12 panel contrasts. Open symbols: panel-balanced means. Bars: 95% panel-cluster bootstrap intervals. Global interaction: $p = 0.414$.

Figure 4 The central empirical comparison. Panels A–B show that raw Pearson predictions gain 0.0048 in correlation but increase normalized RMSE by 0.346 relative to raw MSE; affine calibration retains the correlation gain (+0.0047) and removes the scale penalty (−0.0033). Panel C shows that raw NDCG@10 changes by only +0.0018. Panel D reports estimates and 95% panel-cluster bootstrap intervals for the pooled result and all seven architectures. Small points in A–C are contrasts for the 12 panels.

Validation-fitted affine calibration reduces scale error but is not constrained positive

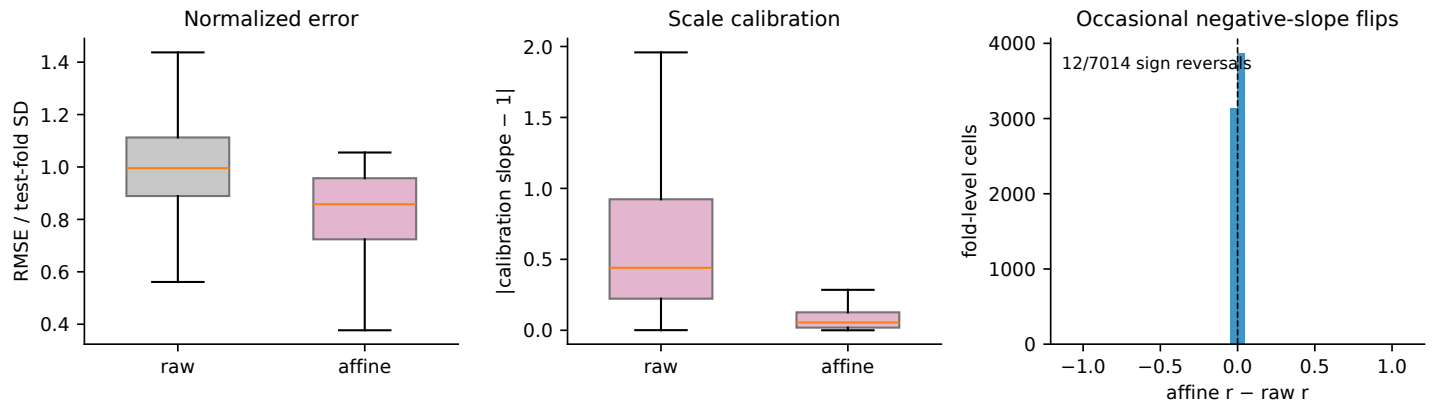


Figure 5 Validation-fitted affine calibration reduces scale error, but an unconstrained negative slope reversed correlation in 12 of 7,014 comparable Pearson-trained fold cells.

Nested-HPO sensitivity: MLP, CNN and Transformer only

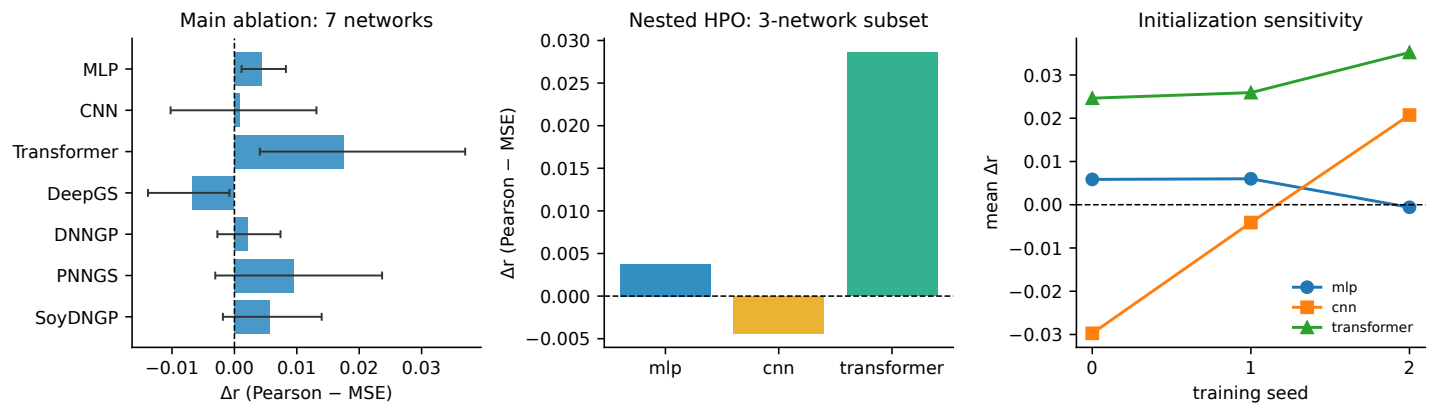


Figure 6 Shared versus loss-specific HPO and three-seed sensitivity. Panel A contains all seven networks. Panels B–C contain only MLP, CNN and Transformer because nested HPO was run only for these three; DeepGS, DNNGP, PNNGS and SoyDNGP were not evaluated in the nested sensitivity. The nested experiment is limited to four outcomes from one CIMMYT wheat panel.

■ **Table 1** The 101 trait–dataset combinations across 12 panels/sources (10 species; 10 panels from EasyGeSe, plus CIMMYT wheat and SoyNAM). *n* is the number of genotypes, *p* the markers used (after MAF filtering and the 20,000-marker variance cap), and GBLUP *r* the mean within-dataset predictive ability of GBLUP (a difficulty index).

Species	<i>n</i>	<i>p</i>	traits	GBLUP <i>r</i>	source
rice	352	20,000	36	0.46	EasyGeSe
pine	925	4,421	17	0.82	EasyGeSe
wheatG	371	12,546	16	0.59	EasyGeSe
maize	4421	20,000	6	0.83	EasyGeSe
bean	444	16,707	5	0.57	EasyGeSe
lentil	324	20,000	5	0.79	EasyGeSe
soybeanNAM	5487	4,401	4	0.71	SoyNAM
wheat	599	1,279	4	0.46	BGLR/CIMMYT
barley	1438	20,000	2	0.58	EasyGeSe
oyster	372	20,000	2	0.38	EasyGeSe
pig	1709	20,000	2	0.31	EasyGeSe
soybean	346	20,000	2	0.40	EasyGeSe

The count of 12 panels/sources (spanning 10 species) treats related germplasm sharing a common crop as distinct panels: wheatG (EasyGeSe) and wheat (CIMMYT) are separate sources, as are soybean (EasyGeSe) and soybeanNAM (SoyNAM) panels of the same species drawn from different sources.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36

Desta, Z. A. and R. Ortiz, 2014 Genomic selection: genome-wide prediction in plant improvement. *Trends in Plant Science* **19**: 592–601.

Endelman, J. B., 2011 Ridge regression and other kernels for genomic selection with R package rrBLUP. *The Plant Genome* **4**: 250–255.

Gao, P., H. Zhao, Z. Luo, Y. Lin, W. Feng, *et al.*, 2023 SoyDNGP: a web-accessible deep learning framework for genomic prediction in soybean breeding. *Briefings in Bioinformatics* **24**: bbad349.

Harris, C. R., K. J. Millman, S. J. van der Walt, *et al.*, 2020 Array programming with NumPy. *Nature* **585**: 357–362.

Heffner, E. L., M. E. Sorrells, and J.-L. Jannink, 2009 Genomic selection for crop improvement. *Crop Science* **49**: 1–12.

Jannink, J.-L., A. J. Lorenz, and H. Iwata, 2010 Genomic selection in plant breeding: from theory to practice. *Briefings in Functional Genomics* **9**: 166–177.

Järvelin, K. and J. Kekäläinen, 2002 Cumulated gain-based evaluation of IR techniques. *ACM Transactions on Information Systems* **20**: 422–446.

Kingma, D. P. and J. Ba, 2015 Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*.

Lin, L. I.-K., 1989 A concordance correlation coefficient to evaluate reproducibility. *Biometrics* **45**: 255–268.

Ma, W., Z. Qiu, J. Song, J. Li, Q. Cheng, *et al.*, 2018 A deep convolutional neural network approach for predicting phenotypes from genotypes. *Planta* **248**: 1307–1318.

Meuwissen, T. H. E., B. J. Hayes, and M. E. Goddard, 2001 Prediction of total genetic value using genome-wide dense marker maps. *Genetics* **157**: 1819–1829.

Montesinos-López, O. A. *et al.*, 2025 Artificial intelligence meets genomic selection: comparing deep learning and GBLUP across diverse plant datasets. *Frontiers in Genetics* **16**: 1568705.

Montesinos-López, O. A., A. Montesinos-López, C. M. Hernandez-Suarez, J. A. Barrón-López, and J. Crossa, 2021a Deep-learning power and perspectives for genomic selection. *The Plant Genome* **14**: e20122.

37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
66
67
68
69
70
71
72

Montesinos-López, O. A., A. Montesinos-López, P. Pérez-Rodríguez, *et al.*, 2021b A review of deep learning applications for genomic selection. *BMC Genomics* **22**: 19.

Ornella, L., P. Pérez, E. Tapia, J. M. González-Camacho, *et al.*, 2014 Genomic-enabled prediction with classification algorithms. *Heredity* **112**: 616–626.

Paszke, A., S. Gross, F. Massa, *et al.*, 2019 PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32.

Pedregosa, F., G. Varoquaux, A. Gramfort, *et al.*, 2011 Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* **12**: 2825–2830.

Pérez, P. and G. de los Campos, 2014 Genome-wide regression and prediction with the BGLR statistical package. *Genetics* **198**: 483–495.

Quesada-Traver, C., D. Ariza-Suarez, B. Studer, and S. Yates, 2025 EasyGeSe: a resource for benchmarking genomic prediction methods. *BMC Genomics* **26**: 953.

VanRaden, P. M., 2008 Efficient methods to compute genomic predictions. *Journal of Dairy Science* **91**: 4414–4423.

Waldmann, P., 2019 On the use of the pearson correlation coefficient for model evaluation in genome-wide prediction. *Frontiers in Genetics* **10**: 899.

Wang, K., M. A. Abid, A. Rasheed, J. Crossa, S. Hearne, *et al.*, 2023 DNNGP, a deep neural network-based method for genomic prediction using multi-omics data in plants. *Molecular Plant* **16**: 279–293.

Washburn, J. D. *et al.*, 2025 Global genotype by environment prediction competition reveals that diverse modeling strategies can deliver satisfactory maize yield estimates. *Genetics* **229**: iyae195.

Xavier, A., W. M. Muir, and K. M. Rainey, 2016 Assessing predictive properties of genome-wide selection in soybeans. *G3: Genes, Genomes, Genetics* **6**: 2611–2616.

Xie, Z., L. Weng, J. He, X. Feng, X. Xu, *et al.*, 2024 PNNGS, a multi-convolutional parallel neural network for genomic selection.

